FAIR 3D HERITAGE
Documentation, Publication and Preservation of the Digital 3D Heritage on the Web

Integration of the fourth dimension in heritage visualisation: A GIS-Based environment as a tool for multi-source data integration

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Abstract

The fourth dimension involves incorporating time into 3D models (Mammoli et al., 2021; Rodríguez-Gonzálvez et al., 2017), which facilitates the visualisation and analysis of the evolution of assets. In the field of heritage, its application is relevant to architecture, archaeology, conservation, museography, and outreach (Hu et al., 2025), and is especially useful in contexts where primary source information is dispersed. Initiatives have effectively developed 4D environments to reconstruct and visualise urban evolution from heterogeneous historical sources (Altes Leipzig, 2026), as well as collaborative platforms that integrate archives and citizen participation for the analysis of urban transformation (Altes Köln, 2019).

The case study belongs to Barranquilla, Colombia, where historical records from the early 20th century are fragmented. For much of that century, building codes were governed by the 1887 Civil Code (Roa Muñoz & Rodríguez Zárate, 2015), which imposed minimal documentation and preservation requirements, while local archives were primarily established in the second half of the 20th century (Asamblea Departamental del Atlántico, 1992; Alcaldía de Barranquilla, 2025). For these reasons, much of the information of this timeframe is scattered, making the use of the fourth dimension crucial for its organisation.

This research proposes a prototype 4D digital environment that integrates multisource data on a single platform, including texts, deeds, newspapers, advertisements, historical urban plans and photographs, integrated with an interpretive 3D model that facilitates the convergence of this information. The selected case examines the demolished Palma Building and its surrounding urban area in Barranquilla.

Its methodology uses ArcGIS Pro to create a four-dimensional timeline, segmented into four eras, thereby enabling the visualisation of the spatial narrative's evolution. Accordingly, the GIS framework acts as the underlying spatial data system that structures and manages all information layers, while the 3D components are the reconstructed building and polygons. Therefore, the 4D environment is the result of combining these through a temporal dimension, where each historical period activates a different state of the model over time.

Historical cartographic sources in raster format are georeferenced and superimposed onto existing GIS datasets stored as geodatabases and shapefiles. Vector spatial features represent reconstructed urban and architectural elements, aligning historical and present-day landscapes. Furthermore, a 3D model of the building created in Autodesk Revit is imported and integrated with attribute-based pop-ups, allowing access to historical evidence and plans indicating the reconstruction confidence levels per facade.
Consequently, the prototype communicates a narrative of demolished heritage and urban transformation processes, serving as a dissemination tool with potential publication via an ArcGIS Web App, and the results (Fig. 1-2) demonstrate the effectiveness of this method in compiling and organising diverse historical evidence in a digital environment, fostering interaction by integrating 3D and 2D resources with textual information. Designed considering FAIR principles, the workflow brings together dispersed materials, improving accessibility and potential reuse, whilst supporting future discoverability by including metadata managed through web-based GIS item descriptions structured according to Dublin Core-compatible fields. Moreover, interoperability relies on standard GIS data formats of ArcGIS and paradata enhances reusability by capturing interpretative decisions and modelling assumptions.

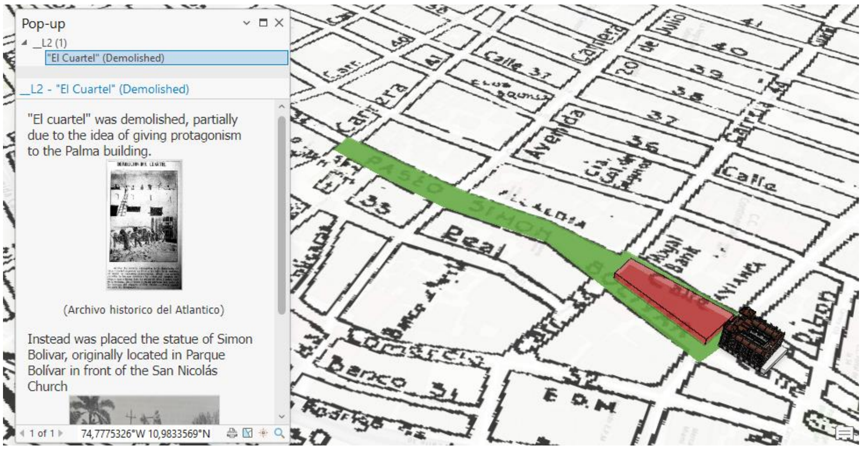

Fig. 1 4D Environment on ArcGIS. Image created by the author using a historical map from Empresas Públicas Municipales and an archival photograph from Archivo Histórico del Atlántico.

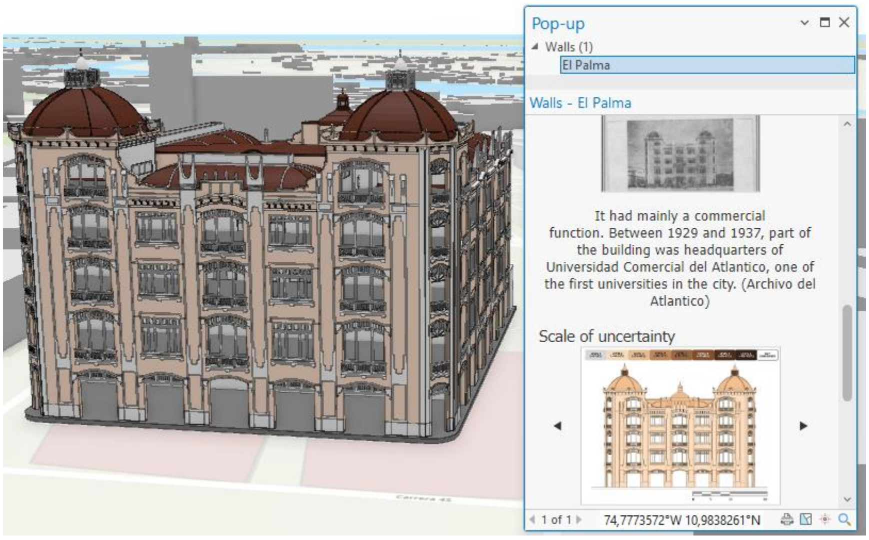

Fig. 2 Hypothetical 3D reconstruction of the Palma building. Image created by the author using an archival photograph from Archivo Histórico del Atlántico.

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